Research Facilities:

Systems Security and Innovation (SSI) Lab

The Systems Security and Innovation (SSI) Lab focuses on cutting-edge research in cybersecurity, secure software systems, blockchain technologies, and privacy-preserving solutions. It supports the design and evaluation of resilient digital infrastructures for Industry 4.0 and smart systems.

IoT and Robotics Lab

The IoT and Robotics Lab develops intelligent, connected systems integrating sensors, embedded platforms, and automation technologies. The lab emphasizes smart environments, industrial automation, and real-time data-driven robotic applications.

AI and Machine Learning Lab

The AI and Machine Learning Lab advances research in data analytics, deep learning, natural language processing, and computer vision. It aims to solve real-world problems through intelligent algorithms and scalable AI-driven solutions.

Bioinformatics and Drug Designing Lab

This lab applies computational techniques to biological data analysis and drug discovery. It focuses on genomics, proteomics, molecular modelling, and in silico drug design to accelerate biomedical research and therapeutic innovation.

Molecular Biology Lab

The Molecular Biology Lab conducts experimental research on cellular and molecular mechanisms of life. It supports studies in genetics, microbiology, and biotechnology, with applications in health, agriculture, and environmental science.

Material Science Lab

The Material Science Lab explores the synthesis, characterization, and application of advanced materials. Research areas include nanomaterials, polymers, and functional materials for use in energy, electronics, and environmental sustainability.

Selected Research Grants:

Mawlana Bhashani Science and Technology University (MBSTU) marks a new milestone in scientific research — 11 research projects approved under the Ministry of Science and Technology, Government of Bangladesh.

Project 1 (ID: 11026)

Title:
Novel Bacteriophage-Based Strategies for the Eradication of Waterborne Microbes Causing Several Epidemics

Affiliation:
Mawlana Bhashani Science and Technology University

Principal Investigator:
Dr. Mohammod Johirul Islam

Department:
Department of Biochemistry and Molecular Biology

Research Focus:
This project explores bacteriophage-based biocontrol mechanisms targeting pathogenic waterborne microorganisms, aiming to develop sustainable, antibiotic-free strategies for epidemic prevention and water safety.

https://old.heat.ugc.gov.bd/notices/45

Project 2 (ID: 12327)

Title:
Transforming Biochemistry and Molecular Biology Education through Cutting-Edge Laboratory and Digital Resources to Meet Global Challenges

Affiliation:
Mawlana Bhashani Science and Technology University

Principal Investigator:
Dr. Md. Asaduzaman Sikder

Department:
Biochemistry and Molecular Biology

Research Focus:
This project emphasizes modernization of laboratory infrastructure and integration of digital learning platforms to enhance competency-based education aligned with global standards in life sciences.

https://old.heat.ugc.gov.bd/notices/45

Project 3 (ID: 12326)

Title:
Improvement of the Teaching-Learning Atmosphere to Produce Qualified Graduate Pharmacists Ready to Address Global Health Challenges

Affiliation:
Mawlana Bhashani Science and Technology University

Principal Investigator:
Professor Dr. Md. Mizanur Rahman Moghal

Department:
Department of Pharmacy

Research Focus:
The project targets pedagogical innovation and curriculum enhancement to produce industry-ready pharmacists equipped to tackle emerging global health issues.

https://old.heat.ugc.gov.bd/notices/45

Project 4 (ID: 13553)

Title:
Development of Low-Cost Electrochemical Sensors for the Determination and Mechanistic Investigation of Extensively Used Antibiotics Residue from Clinical, Environmental, and Food Matrices

Affiliation:
Mawlana Bhashani Science and Technology University

Principal Investigator:
Dr. Md. Abu Rashed

Department:
Department of Chemistry

Research Focus:
This research develops affordable electrochemical sensing technologies for detecting antibiotic residues, addressing antimicrobial resistance and environmental contamination challenges.

https://old.heat.ugc.gov.bd/notices/45

Among the 662 projects selected nationwide under the Science and Technology Research Grant Program 2025–26, 11 projects from MBSTU have been approved.

Notably, four of these belong to the Department of Biochemistry and Molecular Biology, which are listed below:

Nazrul Islam, Nargis Akter, Agri-VoiceLink: A Hybrid Conversational AI Platform to Enhance Agricultural Market Efficiency in Bangladesh, Ref: https://icsetep.ugc.gov.bd/wp-content/uploads/2026/01/ASP-Details.pdf Amount: 2 Crore 45 Lacs.

1. Project No.: SRG-255570

Principal Investigator: Professor Dr. **Mohammod Johirul Islam**

Associate Investigator: Lecturer **Fahim Alam Nobel**

Department: Biochemistry and Molecular Biology

Project Title: Integrating PCR-RFLP and tNGS to Elucidate TP53 codon 72 Polymorphism in Osteoporosis Risk Assessment among Bangladeshi Population.

2. Project No.: SRG-255571

Principal Investigator: Assistant Professor Dr. **Mohammad Nasir Uddin**

Associate Investigator: Lecturer Mohammad kamruzzaman

Department: Biochemistry and Molecular Biology

Project Title: Characterization of deleterious missense SNPs in the LIG3 gene in Acute Myeloid Leukemia Patients in Bangladesh: An Integrated Dry and Wet Lab Study.

3. Project No.: SRG-255572

Principal Investigator: Associate Professor Dr. **Khairul Islam**

Associate Investigator: Associate Professor Dr. Md. **Abdullah Al Mamun**

Department: Biochemistry and Molecular Biology

Project Title: Mechanistic Insights into Arsenic-Induced Metabolic Disorders: Developing Plant-Based Targeted Preventive Therapeutics via In Vivo, In Vitro, and In Silico Approaches in Animal Models.

4. Project No.: SRG-255573

Principal Investigator: Professor Dr. Md. Asaduzzaman Sikder

Associate Investigator: Associate Professor **Marina Merri**

Department: Biochemistry and Molecular Biology

Project Title: Identification of upregulated hub genes in non-small cell lung cancer and the effect of chitosan–TPP encapsulated quercetin nanoparticles (QCS-NPs) on hub genes as a therapeutic candidate: Insights from InSilico and In-Vivo Studies.

Funding Sponsor

Ministry of Science and Technology, Government of the People’s Republic of Bangladesh
Japan Society for the Promotion of Science
Bangladesh Council of Scientific and Industrial Research
University Grants Commission of Bangladesh
Princess Nourah Bint Abdulrahman University
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Singapore
Poland
Philippines
Pakistan
Norway
Nepal
Lithuania
Jordan
Ireland
Iraq

Iran
Hungary
Belgium
Algeria

Selected Interdisciplinary Research

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